SPANDAN **BHOL**

AI ENGINEER

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Profile Summary

- AI Engineer with **2.5+ years** of specialized experience in ***Generative AI***, ***LLM applications***, and ***agentic AI systems***. Genuinely passionate about building AI-based products to solve real-world problems with a focus on practical, production-ready implementations users would love!
- Strong ***DevOps background*** (**2 years**) positioning me well for seamless deployments and scaling of AI systems in cloud environments.
- Overall total experience of 11 years including non-IT experience.

Skills

AI/ML Technologies

Large Language Models (LLMs)	OpenAI GPT, Claude, Llama, Mistral, Cohere, Gemini, Qwen
Agentic LLM Frameworks	LangChain, LangGraph, LlamaIndex, AutoGen, Function Calling, Tool Use
Embeddings & Vector Databases	OpenAI Embeddings, Sentence-Transformers, BGE-M3, FAISS, Pinecone, Weaviate, Chroma, Qdrant, Neo4j, Vespa, S3 Vectors
Prompt Engineering	Zero-shot, Few-shot, Chain-of-Thought, ReAct, Tree-of-Thoughts
RAG Techniques	Hybrid Search, Re-ranking, Query Decomposition, HyDE, Multi-Query Retrieval, Parent-Child Chunking
Transformers & NLP	Hugging Face Transformers, Tokenizers, PEFT (LoRA, QLoRA), BERT, GPT Architecture
Speech/Voice Technologies	Wake Word Detection, Voice Biometrics (ECAPA-TDNN), Speaker Diarization

Programming & Development

Core Languages	Python, Bash, SQL, JavaScript (Basic)
Python Stack/AI/ML Libraries	NumPy, Pandas, Scikit-learn, FastAPI, Pydantic, Asyncio, PyTorch, Gradio

MLOps/LLMOps

Model Serving	vLLM, TGI, Ray Serve, LiteLLM | GPU optimization, batching
Monitoring	Prometheus, Grafana, LangSmith, Weights & Biases
Experiment Tracking	MLflow, Weights & Biases, LangSmith tracing & evaluation
Orchestration	Airflow, Prefect | Cost tracking, A/B testing, prompt versioning

Skills (continued)

DevOps & Cloud	
Cloud Platforms	AWS - EC2, Lambda, S3, SageMaker, Bedrock, ECS Azure - APIM, Azure OpenAI GCP - Vertex AI, Google AI Studio, Firebase, Firestore Vast.ai - GPU compute for model deployment
CI/CD	Jenkins, GitHub Actions, Bitbucket Pipelines
Infrastructure as Code	Terraform, CloudFormation
Version Control	GitHub, BitBucket
Databases	PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, SQLite
Third-Party Integrations	Gmail, Google Drive, Dropbox, JIRA, Outlook Mail, Slack, Telegram
Other Tools	Postman, Docker Compose, Jupyter Notebook, Cursor, Windsurf, Kiro, VS Code, lightning.ai

Work Experience

AI Engineer | Feb 2025 - Oct 2025 | Huemn Interactive Pvt. Ltd. (ve.ai)

Project 1: Vector DB & Embedding Evaluation

Duration: 1.5 months | Team Size: 2

- Evaluated 5 vector DBs (Pinecone, Weaviate, Neo4j, Vespa, S3) on 1M+ vectors
- Tested embedding models (BGE-M3, OpenAI, Sentence-Transformers) for production
- Built automated benchmarking pipeline: Docker containers for each DB, standardized tests
- Monitoring: Tracked query latency (p50, p95, p99), throughput, memory usage, cost per query
- Created comparison dashboard with Grafana visualizing performance metrics across DBs
- Finalized BGE-M3 (faster, cheaper) + Pinecone (best performance)
- **Tech**: BGE-M3, Pinecone, Weaviate, Docker, Redis, Prometheus, Grafana, Python

Project 2: Modular Agent Tools & Third-Party Integrations

Duration: 2.5 months | Team Size: 3

- Built modular tools (web search, file parsing, brand voice & persona generator) for the ve.ai pipeline.
- Integrated third-party services (Gmail, Google Drive, Dropbox, JIRA, Outlook, Slack, Notion) with OAuth 2.0 into the ve.ai pipeline for the VE Agentic Assistant to have comprehensive capabilities.
- **Tech**: LangChain, Tavily API, GPT-4, Gmail/Drive/Dropbox/JIRA APIs, FastAPI, PyPDF2, python-docx

Project 3: Voice AI & Agent Systems

Duration: 2 months | Team Size: 2

- Evaluated/deployed self-hosted STT models (Nvidia Parakeet, Kyutai) on Vast.ai GPU
- Built WebSocket streaming for real-time transcription (<500ms latency)
- Integrated OpenWakeWord + ECAPA-TDNN for wake word + voice biometrics (98% accuracy)
- Model Serving: Optimized PyTorch models with ONNX quantization (2x faster inference)
- Monitoring: Implemented latency tracking, error rate monitoring, cost per transcription
- **Tech**: OpenWakeWord, ECAPA-TDNN, Vast.ai, PyTorch, FastAPI, WebSockets, ONNX, Docker, Node.js, PyAudio, Electron

Project 4: Browser Agent Development

Duration: 2 months | Team Size: 4

- Built intelligent browser agent using Eko framework + Chrome DevTools Protocol (CDP)
- Implemented multi-step reasoning with LangGraph for task decomposition
- Created visual understanding with GPT-4 Vision for screenshot-based navigation
- **Tech**: LangGraph, GPT-4 Vision, Eko Framework, CDP, Python, FastAPI, Docker, WebSockets

AI Engineer | Apr 2023 - Feb 2025 | Tritech Software Pvt. Ltd.

Project 1: Conversational SQL Agent for Business Intelligence

Duration: 2.5 months | Team Size: 4 | Client: SaaS Analytics Company

- Built NL-to-SQL agent with few-shot prompting + semantic example selection (88% accuracy)
- Created safety layer preventing destructive ops + PII exposure
- Designed Redis caching reducing API costs 60%, response time 75%
- Impact: 50+ non-tech users enabled, 40% less analyst workload, $2K/month saved
- **Tech**: LangChain, GPT-4, SQLAlchemy, Sentence-Transformers, Redis, PostgreSQL

Project 2: Autonomous Research Agent for Consulting Firm

Duration: 4.5 months | Team Size: 3 | Client: Management Consulting Firm

- Built multi-tool agent with 12 tools: web search, PDF parsing, data analysis, email
- Implemented ReAct pattern + human-in-the-loop for sensitive operations
- Created memory system with conversation history + entity tracking
- Impact: 70% faster research, 500+ requests, 200+ reports, 150+ hours/month saved
- **Tech**: LangChain, LangGraph, GPT-4, Tavily, Redis, PostgreSQL, AWS Lambda

Project 3: Enterprise RAG System for Customer Support

Duration: 6 months | Team Size: 3 | Client: US-based Retail Company

- Built production RAG serving 5K+ daily users with 92% accuracy
- Implemented LangGraph multi-agent: router → retriever → synthesizer → validator
- Designed hybrid retrieval (BM25 + embeddings) + Cohere re-ranking (+35% accuracy)
- Processed 100K+ documents with semantic chunking + metadata extraction
- MLOps Infrastructure: Containerized with Docker, deployed on AWS ECS with auto-scaling
- Monitoring: Prometheus + Grafana dashboards tracking latency (p95 <200ms), cost, quality
- Experiment Tracking: LangSmith for prompt versioning, A/B testing, evaluation metrics
- Cost Optimization: Redis caching + batching reduced API costs 40%, response time 60%
- **Tech**: Langchain, LlamaIndex, FAISS, GPT-4, Claude, Redis, Prometheus, Grafana, PostgreSQL

Project 4: Document Q&A System for Legal Compliance

Duration: 4 months | Team Size: 2 | Client: European Legal Services Firm

- Built RAG with precise citations using parent-child chunking (88% citation accuracy)
- Optimized retrieval with HyDE, query expansion, multi-query techniques (+30% quality)
- Implemented RAGAS evaluation pipeline with real-time monitoring
- Impact: 50% less manual review time, 10K+ queries processed, 4.2/5 satisfaction
- **Tech**: LlamaIndex, Pinecone, Claude 3.5, GPT-4, Sentence-Transformers, AWS S3

Project 5: Meeting Intelligence System

Duration: 4 months | Team Size: 4 | Client: Enterprise Software Company

- Built meeting intelligence with real-time transcription + speaker diarization
- Created agent for live insights: action items, decisions, key points
- Implemented searchable archive with semantic search across all meetings
- Production Deployment: Containerized with Docker, deployed on AWS with load balancing
- Monitoring: Real-time dashboards tracking transcription accuracy, latency, error rates
- Cost Optimization: Hybrid approach (Whisper for batch, AssemblyAI for real-time) saved 50%
- **Tech**: AssemblyAI, Whisper, LangChain, GPT-4, MongoDB, Pinecone, Docker, AWS, Prometheus

Project 6: Enterprise LLM Gateway with MLOps Infrastructure

Duration: 2 months | Team Size: 1 | Client: Multi-National Tech Company

- Built production LLM gateway using Azure APIM using precise policy for flawless execution
- Implemented multi-account Azure OpenAI strategy with auto-rotation
- Created rate limiting, quota management, cost tracking per department
- Finalized on Azure APIM instead of AWS Lambda-based solution after experimenting extensively on the latter.
- Optimized costs by migrating from Basic tier to Consumption tier after thorough evaluation.
- Impact: $5K/month saved, 99.9% uptime
- **Tech**: Azure APIM, Azure OpenAI, AWS Lambda, Powershell 7, Azure Monitor

Devops Engineer | July 2021 - Mar 2023 | Tritech Software Pvt. Ltd.

- Worked on multiple client projects involving cloud migration, infrastructure automation, security implementation, and monitoring solutions, hence gaining a thorough understanding of the SDLC.
- Developed expertise in AWS services, Docker containerization, Terraform infrastructure provisioning, and CI/CD pipeline automation.
- This DevOps foundation proved invaluable for deploying production AI applications with high availability, security, scalability, and cost-efficiency in mind.

Education

B.Tech (Electrical)	2008 - 2012	G.I.E.T., Gunupur	7.64/10
Higher Secondary	2005 - 2007	Glorious Academy, Bhubaneswar	68%
Matriculation	2005	V.E.M. School, Bhubaneswar	88.3%

Personal Projects

Likhona - Real-time Voice Transcription Application (Released for Windows)

- Production app with multi-provider STT + multi-provider-LLM + auto-failover designed for leveraging free tier and trial credits. A complete stack with Gen-AI focused models including embeddings and reranking models.
- Wake word activation (ONNX), multi-account switchover, cost tracking, LLM-based correction, rephrasing, or formatting.

Personal Projects (continued)

- Ultra-low latency with WebSocket streaming, auto-disconnect based on VAD for cost savings, ultra-efficient for low resource-usage.
- **Tech**: Python, Deepgram, AssemblyAI, PySide6, Tkinter, PyInstaller, Rust, ONNX, WebSockets

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The AI Radar *- Daily AI News Podcast (Live on Spotify)*

- Fully automated engine for daily AI/ML podcast using agentic workflow with LangGraph. Automated distribution via RSS feed.
- Implemented two episode types: Industry news and Research papers scraping hundreds of data sources.
- Built multi-agent (13) system with 4-host system with specialized personas, cost tracking, Telegram notifications, and episode tracking for production monitoring.
- Implemented memory layer (SQLite + FAISS) for accumulating AI developments over time.
- Implemented broker lambdas which handle Deepgram and Tavily accounts rotated seamlessly, leveraging a pool of free tier/free credit accounts.
- **Tech:** Python, LangGraph, OpenAI API, Deepgram TTS, AWS, SQLite, Tavily, RSS.

Jugglerr *- Multi-Provider LLM Routing Library (Open Source, published on PyPI)*

- Python library for intelligent routing across multiple LLM providers.
- Implemented multi-key rotation system to maximize free-tier usage across providers with automatic key cycling among multiple accounts and automatic fallbacks.
- Designed capability-based routing engine matching requests to optimal models based on vision, tool calling, context window, reasoning and raw power (parameter size) requirements.
- Supports specialized model adapters for safety/moderation, embeddings, reranking and multi-language translation.
- **Tech**: Python, Groq/Cerebras/Mistral/NVIDIA NIM/Cohere APIs, PyPI

IMDb Magic *– Browser Extension facilitating free streaming (released on Mozilla Firefox)*

- A JavaScript-based tool that adds an interactive button to IMDb pages for direct streaming access.
- DOM manipulation to create an interactive UI with animations and overlays for seamless navigation.
- A (deliberately) not-so-popular extension for users' convenience.
- **Tech:** Firefox WebExtensions API, *Plain old* Javascript!

General Info

Personal — Bhubaneswar, Odisha — English, Hindi, Odia

Interests — Movies/Series — Animal-lover — Gaming — Table Tennis